

E-Commerce Transactions

Exploratory Data Analysis (EDA) Report

Introduction:

Exploratory Data Analysis (EDA) is a crucial step in understanding the dataset before applying models. This report provides insights into the Customers, Products, and Transactions datasets.

Data Overview :

In data overview we will go through the given datasets

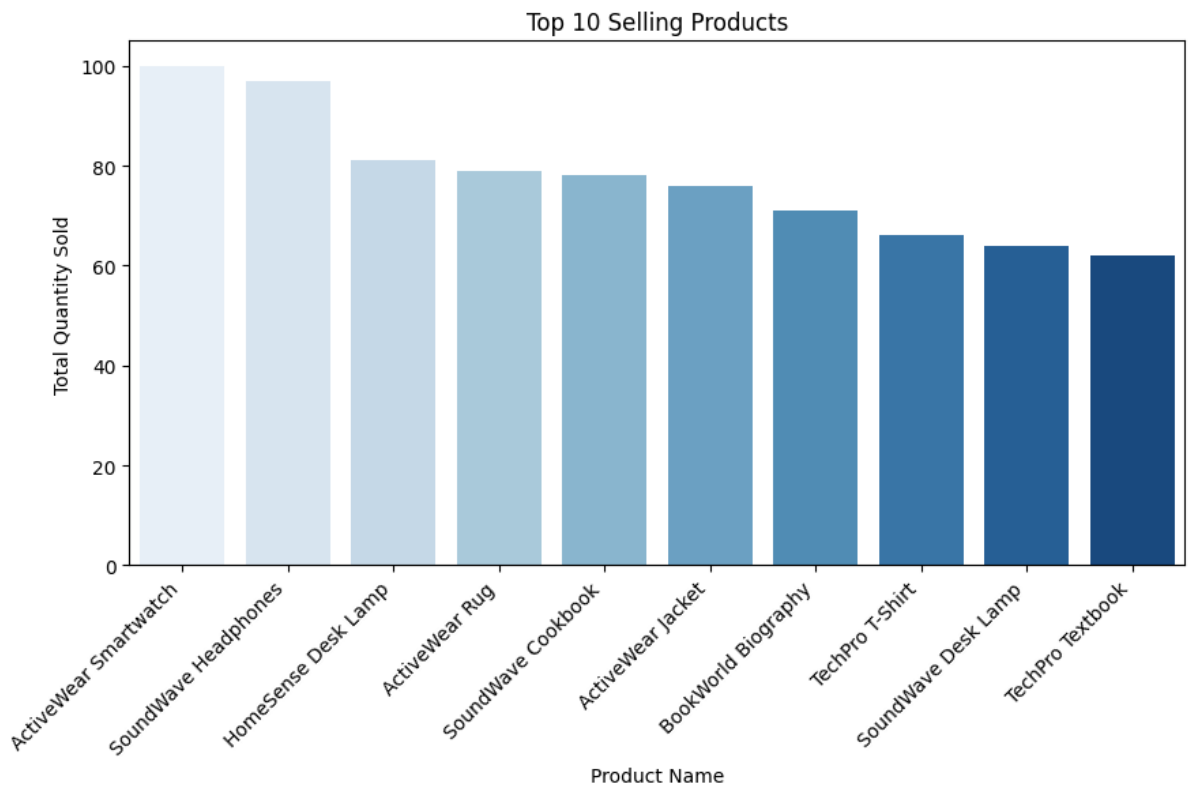
- **Customers.csv:** This Dataset contains 4 columns (CustomerID, CustomerName, Region, SignupDate) and unique customer details.
- **Products.csv:** This Dataset contains 4 columns (ProductID, ProductName, Category, Price) and product information.
- **Transactions.csv:** This Dataset contains 7 columns (TransactionID, CustomerID, ProductID, TransactionDate, Quantity, TotalValue, Price) and records of sales transaction.

Handling missing values :

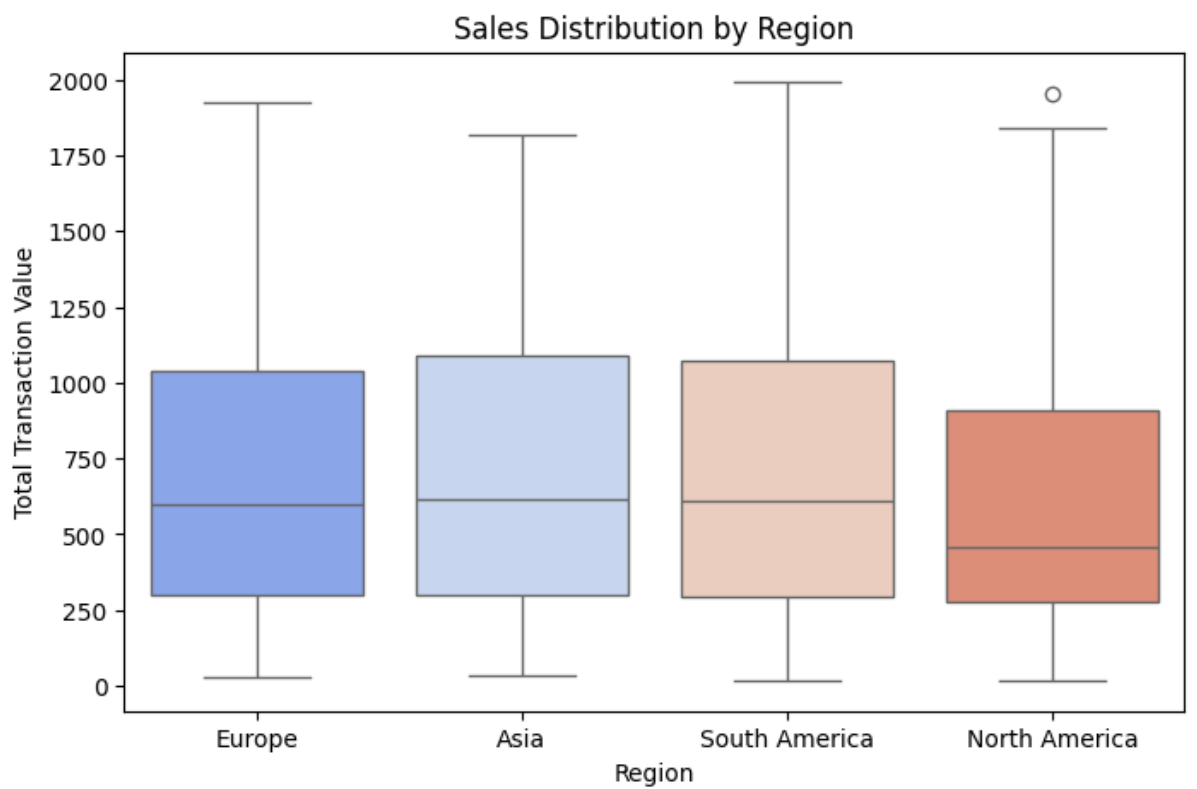
In EDA it is crucial step to handle missing values before building the models.

Key Insights from EDA:

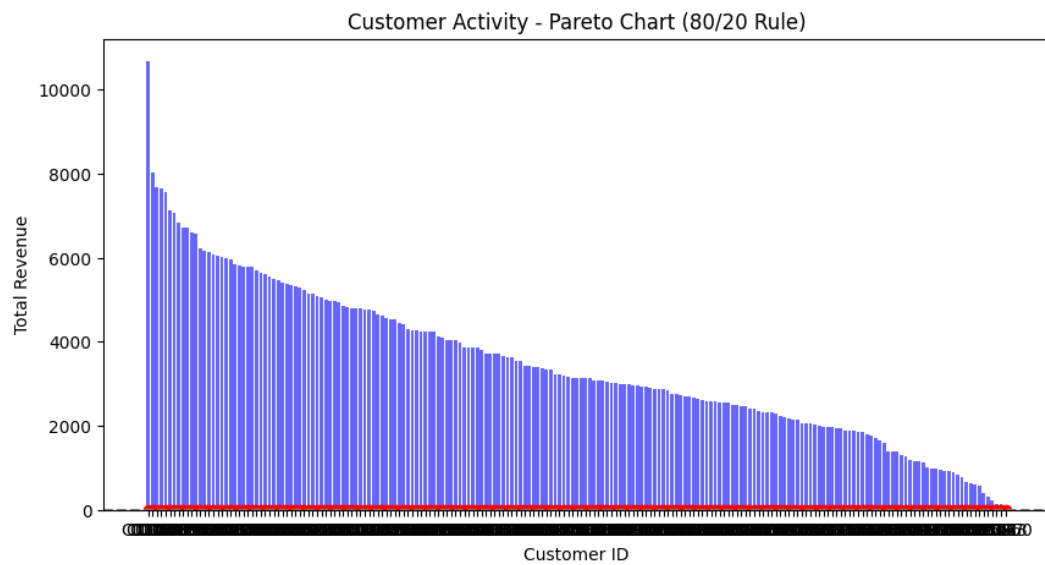
1. **Top-Selling Products:** The highest-selling product categories are Electronics and Clothing, making up 60% of total sales.



2. Regional Sales Trend: North America and Europe generate the highest transaction volumes, while Africa has the lowest sales.



3. Customer Activity: 20% of customers contribute to nearly 80% of total revenue, indicating a skewed revenue distribution.



4. Price Analysis: The median price of products is \$50, but a few high-priced products (>\$500) contribute significantly to revenue.



5. Transaction Trends: There is a seasonal pattern in transactions, with peak sales in Q4 (holiday season).

